

Daily Paper Review: From RLVR to RLSVR — Task Transformation Induces Self-Verifiable Rewards for Open-Ended LLM Self-Improvement

Automated Research Review

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1 Paper Summary

1.1 Problem and Motivation

Reinforcement learning with verifiable rewards (RLVR) has driven substantial reasoning gains in domains like mathematics and coding where correctness is deterministically checkable. However, RLVR is **fundamentally inapplicable to open-ended tasks**—creative writing, summarization, general language generation—because these tasks lack deterministic verification criteria. Existing alternatives each have shortcomings: RLHF/DPO suffer from preference data cost and reward model staleness; LLM-as-a-Judge introduces capability bottlenecks and evaluation bias; Rubric-as-Reward (RaR) incurs significant verifier costs (\$200–\$900 per training run). Self-play frameworks like R-Zero and Absolute Zero rely on task verifiability and deteriorate on subjective tasks.

1.2 Method

RLSVR [1] introduces a paradigm shift: rather than building better reward models for unverifiable tasks, it **transforms the task itself** into a proxy environment where rewards become automatically verifiable through environmental rules.

The RLSVR Framework operates in four steps: (1) the environment samples input x and records a latent variable z as ground truth, (2) observations constructed from (x, z) are given to the policy for task execution, (3) environmental rules pose questions about z answerable only from task outputs, and (4) rule-based rewards are computed by checking against z , requiring no human annotation, learned reward model, or external judge.

SpyRL concretely instantiates RLSVR through an information-asymmetric multi-agent game. Among $n = 5$ players, one is randomly designated as the “spy” (index u), receiving a *degraded* input $o_u = g(x)$ while civilians receive the full input $o_i = x$. The degradation operator $g(\cdot)$ is task-specific: 20% continuous span masking for text summarization/creative writing, 40% partial context removal for mathematical reasoning. All players generate outputs $y_i \sim \pi_\theta^P(\cdot | o_i, \tau)$, then each votes $v_i \in \{1, \dots, n\}$ to identify the spy.

Self-Verifiable Rewards. Since the spy identity u is explicitly assigned by the environment, detection rewards are computed via simple indicator functions: $r_i^D = \mathbf{1}[v_i = u]$. The performing stage uses a **zero-sum reward**:

$$r_u^P = -\beta \cdot (m_u - \bar{m}_c), \quad r_{c_j}^P = \frac{\beta}{n_c} (m_u - \bar{m}_c) - \lambda (m_{c_j} - \bar{m}_c) \quad (1)$$

where m_u is the vote count the spy receives, $\bar{m}_c$ is the civilian average, β controls competitive signal strength, and λ adjusts intra-group consistency pressure. The zero-sum constraint $r_u^P +$

$\sum r_{c_j}^P = 0$ enables continual co-evolution: the spy is pushed to improve (avoid detection), civilians maintain quality advantage, and the λ term prevents free-riding.

Both stages are optimized via PPO-style clipped objectives with KL regularization, updated in **strict alternation**—performing then detection—to reduce credit assignment interference.

1.3 Results

Across summarization, creative writing, and mathematical reasoning on Qwen3-4B and Qwen3-8B:

- **Summarization:** +4.56 ROUGE-L average (Qwen3-4B), 73.92% average win rate over base model across 5 benchmarks. R-Zero shows negligible gains; Absolute Zero reaches 62.8% average win rate.
- **Creative writing:** 84.3% novelty win rate, 81.3% overall win rate on WritingPrompts (Qwen3-4B). R-Zero and Absolute Zero achieve only 48–58% range.
- **Mathematics:** +8.97% average accuracy across 7 benchmarks (Qwen3-4B). Strongest gains on hardest benchmarks: AIME25 +13.3pp (6.7% $\rightarrow$ 20.0%).
- **vs. RaR:** SpyRL achieves 59.3% win rate against Qwen3.5-27B-RaR and 48.9% against GPT-4o-RaR on creative writing—roughly competitive at **zero verifier cost**.
- **Human evaluation:** 10 PhD students confirm 80.0% overall win rate on WritingPrompts vs. base model, consistent with automatic metrics.

Ablations demonstrate all components are critical: “Only Performing” (frozen detector) plateaus at 72.3% Math500 vs. 79.5% for full SpyRL (+7.2pp gap); “Without spy” (no information asymmetry) stagnates at 71.6%; optimal group size is $n = 5$. Reward–quality correlation validated: players receiving more votes consistently produce lower-quality outputs (confirmed by GPT-4o ranking over 100 games).

1.4 Limitations

SpyRL requires $n = 5$ outputs per training step ($5\times$ inference cost vs. standard RL). The degradation operator $g(\cdot)$ must be manually designed per domain. Only two model sizes (4B, 8B) and three task domains are evaluated. Against the strongest baseline (GPT-4o-RaR), SpyRL achieves only $\sim 49\%$ win rate on creative writing, suggesting the quality ceiling may be bounded by the information-asymmetry mechanism.

1.5 Relevance to Research Topics

RLSVR directly advances **T2 (RL/reward modeling)** by providing a fundamentally new approach to reward design for open-ended domains. The task-transformation paradigm—converting unverifiable tasks into verifiable proxy environments—is conceptually analogous to self-supervised learning’s relationship to supervised learning. The zero-sum co-evolutionary dynamics connect to Skill Self-Play’s [3] bi-level optimization, while the information-asymmetry mechanism provides a novel form of curriculum that could complement CoRT’s [4] counterfactual contrast for token-level credit assignment.

2 Follow-up Research Directions

2.1 Direction 1: Self-Verifiable Rewards for On-Policy Distillation in Open-Ended Domains

Gap. Flux-OPD [2] extended OPD to open-ended domains by conditioning teacher models on evolving contexts, but still relies on teacher supervision quality as the implicit reward signal. RLSVR’s spy game provides a teacher-independent, self-verifiable reward that could replace or augment teacher-conditioned targets. Currently, no OPD framework uses self-verifiable environmental rewards to guide distillation.

Approach. Construct a hybrid “SpyOPD” pipeline: (1) the student generates n rollouts per prompt, one with degraded context (playing the “spy”), (2) the spy game’s detection reward provides a verifiable quality signal that selects which student trajectories to distill from the teacher, and (3) Flux-OPD’s conflict-aware contextual correction is applied only to trajectories where the spy game confirms high quality divergence. The zero-sum performing reward replaces external rubric evaluators, while the teacher provides dense token-level supervision for the selected trajectories.

Feasibility. SpyRL’s $5\times$ generation overhead is already comparable to GRPO’s multi-sample rollout (typically 8–16 samples). The spy game can run on the student model alone (no teacher calls for reward), reducing overall cost compared to RaR-based approaches. Expected to enable OPD in creative and subjective domains where Flux-OPD currently depends on teacher-defined contexts.

2.2 Direction 2: Information-Asymmetric Co-Evolution for Diffusion LLM Quality Signals

Gap. Diffusion LLMs trained with RL (V-GRPO [5], Flow-DPPO [6]) rely on verifiable rewards limited to math/code domains. For open-ended text generation—where dLLMs’ bidirectional generation could excel—no self-verifiable reward mechanism exists. The denoising process itself creates a natural information asymmetry: partially denoised outputs contain degraded information compared to fully denoised ones.

Approach. Exploit the denoising trajectory as a natural spy game: at each noise level τ , the dLLM generates outputs from partially corrupted states (analogous to the spy’s degraded input). A detector network evaluates whether outputs at different noise levels are distinguishable from fully denoised outputs. The detection difficulty serves as a self-verifiable proxy for denoising quality. Combine with subliminal clocks’ [7] latent time representations to condition the detector on the model’s internal denoising progress, creating a noise-level-aware quality signal.

Feasibility. The denoising trajectory already provides multiple “degradation levels” for free, eliminating the need for manual $g(\cdot)$ design. The detector can share parameters with the dLLM itself (via a classification head on the final hidden states), adding minimal overhead. V-GRPO’s per-step reward infrastructure directly supports step-level quality signals from the detection game.

2.3 Direction 3: Spy-Game Curriculum for Skill Library Quality Assessment

Gap. Skill Self-Play [3] uses a gated curriculum reward $R_{\text{propose}} = \mathbf{1}\{\text{valid}\} \cdot (1 - 2|v_{\text{solve}} - 0.5|)$ that depends on solve rates from a single solver. This creates a single-point quality estimate vulnerable to solver-specific biases. RLSVR’s multi-agent detection provides a more robust quality signal through collective voting, but has not been applied to skill library curation.

Approach. Replace Skill Self-Play’s single-solver evaluation with a spy-game assessment: for each proposed skill, n agents attempt the skill’s challenge, with one agent receiving a degraded skill description (the “spy”). The detection difficulty measures whether the skill creates meaningful differentiation—skills where the spy is easily detected are too easy, while skills where

detection fails are too hard or poorly specified. This provides a self-verifiable proxy for skill difficulty calibration without external evaluation. The zero-sum dynamics naturally co-evolve the skill library’s difficulty alongside agent capabilities.

Feasibility. The spy game assessment adds n forward passes per skill evaluation, comparable to Skill Self-Play’s existing dual-stream (skill + exploration) overhead. The information degradation operator for skills can simply mask portions of the skill description, requiring no task-specific design. Expected to improve skill library quality by replacing single-point solve rates with robust multi-agent consensus signals.

3 Conclusion

RLSVR introduces a paradigm shift in reward design for open-ended RL: instead of approximating unverifiable objectives with learned reward models, it transforms the task into a verifiable proxy environment. SpyRL’s concrete instantiation—an information-asymmetric multi-agent game where spy identity detection serves as a self-verifiable proxy for output quality—achieves strong gains across summarization (+4.56 ROUGE-L), creative writing (84.3% novelty win rate), and mathematics (+8.97% accuracy) at zero verifier cost. The three follow-up directions extend this paradigm to on-policy distillation in open-ended domains (replacing rubric-based rewards with spy-game verification), diffusion LLM training (exploiting denoising trajectories as natural information asymmetry), and skill library quality assessment (multi-agent detection as robust difficulty calibration).

References

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